

Merchant Payment Product Setup

MOP catalog and network of record, platform presentment currencies, merchant currency policy, and where each validation runs: at setup or in the transaction flow.

4

money-path currencies, plus reporting and economic as derived views

~40

launch currencies across three operational maturity tiers

2

control axes: who sets presentment, who sets the downstream path

73

requirements with testable acceptance criteria

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Summary and design principles

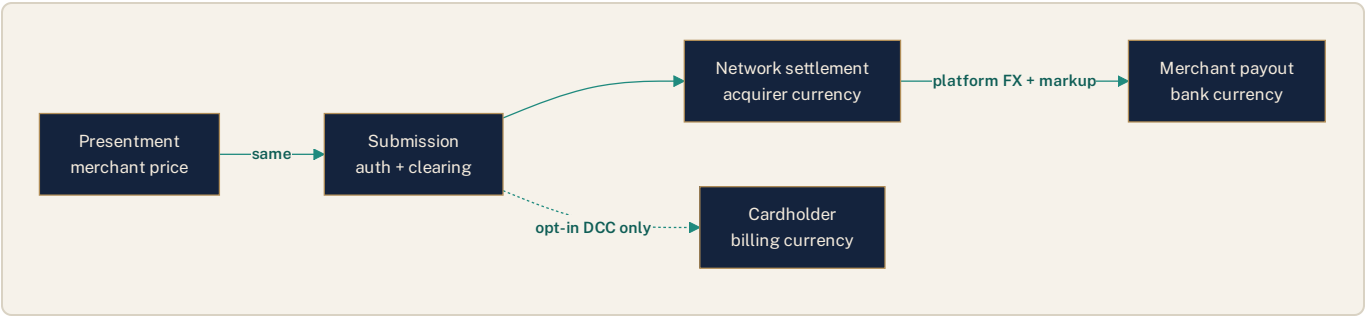
A merchant's payment product is a set of enabled MOPs, each bound to a processing network and a currency capability set. The platform presents and submits in the merchant's priced currency and earns FX only between network settlement and merchant payout.

- 1. Presentment currency = submission currency.** The currency the shopper sees is the currency sent to the network in auth and clearing. The platform never silently converts before submission. Cardholder-side conversion exists only as a separate, opt-in DCC product.
- 2. FX is a settlement-side product.** Conversion happens once, from network settlement currency to merchant payout currency, at a disclosed platform rate plus markup recorded on the ledger entry.
- 3. Capability is a graph, not a list.** The question is not whether the platform supports EUR but how money moves from this shopper to this merchant's bank account: entity, MOP, network, presentment, settlement, payout and the FX corridor between them, each with effective dates.
- 4. Merchant policy narrows, never widens.** Platform capability is the ceiling. A merchant currency policy can only restrict it; the effective set at checkout is $\text{capability} \cap \text{policy}$.
- 5. Policy governs, strategy optimizes.** The policy says which currencies a merchant may use; the strategy says how the platform should use them. They are separate objects because governance changes rarely and strategy is tuned often.
- 6. Currency is a constraint first and an optimization variable second.** Eligibility filters the candidate set; inside what remains the platform optimizes the downstream path — acquiring entity, network, settlement and FX timing — against one economic objective, never by changing what the shopper agreed to pay.
- 7. Validate early, re-validate cheaply.** Structural and contractual checks run at merchant setup; per-transaction checks run against a cached, versioned capability snapshot so auth latency is not affected.
- 8. Same-currency lifecycle.** Captures, refunds, reversals and dispute responses stay in the original transaction currency; any FX difference is posted as an explicit exposure entry, never absorbed and never passed to the shopper.
- 9. Money-path currencies are recorded; analytical currencies are derived.** Reporting and economic currencies are views computed from the ledger. They never rewrite it, and the economic currency never reaches a merchant-facing amount.

Currency model

Four currencies are modeled separately, and only the settlement → payout hop carries platform FX. Stripe draws the same line: it separates the shopper's payment-method currency, the presentment (charge) currency and the settlement currency, and converts only when charge and settlement currency differ ([Stripe](#)).

Currency	Who sets it	Where it lives	Rule
Presentment	Merchant (price)	Order, Session	Must be in the merchant's effective presentment set (capability ∩ policy) for the chosen MOP
Transaction (submission)	Platform, derived	Payment, Attempt	Equals presentment. Differs only on a DCC Attempt, with the DCC quote attached
Settlement	Network + acquiring entity config	Settlement batch, ledger	Fixed per MOP × acquiring entity × transaction currency; not merchant-editable
Payout	Merchant (bank account)	Payout, merchant balance	One default plus optional per-currency bank accounts



The solid path is the default. The dotted DCC path is a separate product with its own quote, disclosure and consent objects.

Amount representation. Store amounts as integer minor units plus an explicit exponent on every money object, not a global lookup. PSPs deviate from ISO 4217: Adyen uses 2 decimals for CLP and ISK and 0 for CVE and IDR, and its own table overrides ISO ([Adyen](#)). Stripe represents ISK and UGX as two-decimal values whose fractional part must be 00 ([Stripe](#)). The platform keeps an ISO exponent, an API exponent and a per-network wire exponent, and converts at the network adapter boundary.

Merchant payment product setup: MOPs and processing networks

The setup assumes a US-domiciled merchant on a US acquiring entity, with optional EU/UK and APAC entities added as the platform expands. Each enabled MOP resolves to exactly one processing network per transaction, chosen by BIN/token attributes and routing policy.

MOP catalog and network of record

MOP	Variants	Processing network / rail	Acquiring model	Presentment scope	Settlement to platform
Visa	Credit, debit, prepaid, commercial	VisaNet (VIP auth, BASE II clearing)	Platform as acquirer	Broad (Tier 1–2)	Visa settlement currency per acquiring BIN
Mastercard	Credit, debit, Maestro, commercial	Banknet auth, GCMS clearing	Platform as acquirer	Broad (Tier 1–2)	MC settlement currency per ICA
US debit (dual-message or PINless)	Visa/MC debit with unaffiliated network	Star, NYCE, Pulse, Accel, Shazam, or the brand, per Reg II routing	Platform as acquirer	USD only	USD
American Express	Consumer, corporate	Amex network	OptBlue (platform settles) or direct SE (Amex settles merchant)	Narrower; Amex excludes some currencies other networks accept	Per Amex SE contract
Discover	Credit, debit	Discover Global Network	Platform as acquirer	USD core, limited others	USD
Diners Club	Consumer, corporate	Discover Global Network (in US)	Platform as acquirer	Same as Discover	USD
JCB	Credit	Discover network for US merchants; JCB International outside US	Platform as acquirer	USD in US; JPY and others via non-US entity	USD in US
UnionPay	Credit, debit	Discover network for US merchants; UnionPay International outside US	Platform as acquirer	USD in US	USD in US
Apple Pay, Google Pay, Click to Pay	Network tokens (DPAN)	Underlying card network via VTS, MDES, Amex Token Service, Discover DTS	Inherits card MOP	Inherits card MOP	Inherits card MOP
Co-badged domestic schemes	Cartes Bancaires, eftpos, Interac	Domestic scheme or co-badged brand, per shopper choice rules	Local acquiring entity required	Domestic currency only	Domestic currency
ACH debit	Web, CCD, PPD	Nacha via ODFI	Platform as originator	USD only	USD

MOP	Variants	Processing network / rail	Acquiring model	Presentment scope	Settlement to platform
Instant bank payments	RTP, FedNow, pay-by-bank	TCH RTP, FedNow	Platform as participant or via sponsor	USD only	USD
SEPA	Direct debit, credit transfer	SEPA schemes via EU entity	EU entity	EUR only	EUR
Local APMs	iDEAL/Wero, Bancontact, BLIK, Pix, UPI, Alipay+, WeChat Pay	Scheme or partner gateway	Partner-acquired, platform as aggregator	Scheme's domestic currency, plus cross-border currencies the partner supports	Partner's settlement currency
BNPL	Klarna, Affirm, Afterpay	Provider's own rails	Provider underwrites, platform routes	Provider's supported markets	Provider's settlement currency

Network-of-record rule. For co-badged and dual-network cards, the network is decided per Attempt, not at setup. Setup only declares which networks the merchant is eligible for. US debit routing must honor Reg II unaffiliated-network availability, and EU co-badge must honor shopper choice under IFR Article 8.

Merchant payment product configuration

```
{
  "merchant_id": "mer_123",
  "legal_entity_country": "US",
  "acquiring_entity": "jpmc_us_01",
  "mcc": "5732",
  "payout_accounts": [
    {"currency": "USD", "default": true},
    {"currency": "EUR", "default": false}
  ],
  "mops": [
    {
      "mop": "visa",
      "variants": ["credit", "debit", "prepaid"],
      "networks_eligible": ["visanet", "star", "nyce", "pulse", "accel"],
      "capture_mode": "auth_then_capture",
      "network_tokens": true,
      "dcc": {"enabled": false}
    },
    {"mop": "amex", "acquiring_model": "optblue"},
    {"mop": "jcb", "networks_eligible": ["discover"]}
  ],
  "fx": {
    "payout_conversion": "platform_rate",
    "markup_bps_tier1": 100,
    "markup_bps_tier2": 200,
    "rate_lock": "at_capture"
  },
  "currency_policy_id": "pol_mer_123_v4",
  "capability_snapshot_version": "cap_2026_09_21_001"
}
```

The merchant configuration does not list presentment currencies per MOP. What the platform *can* process comes from the capability matrix; what the merchant *wants* to accept lives in a separate currency policy, described next.

The markup values are placeholders. The two-tier structure mirrors Adyen's default FX fee, which separates a group of major currencies (AUD, CAD, DKK, EUR, GBP, HKD, JPY, NOK, NZD, PLN, SEK, USD, ZAR) from all others ([Adyen](#)).

Merchant currency policy

Presentment currencies are governed by two layers: the platform capability sets the ceiling, and the merchant policy can only narrow it. The effective set for any Session is capability $\cap$ policy, evaluated per MOP, channel and storefront.

Layer	Owner	Can it add currencies?	Examples
Platform capability	Platform (network certification, acquiring config, ledger, compliance)	Yes, the only layer that can	Tier membership, excluded currencies, MOP $\times$ network $\times$ entity support
Partner or platform-customer policy	ISV or marketplace operating sub-merchants	No, restrict only	Marketplace allows only USD, CAD, EUR for all sellers
Merchant policy	Merchant	No, restrict only	Accept EUR and GBP online, USD only at POS; no JPY on Amex
Storefront or profile policy	Merchant, per site or channel	No, restrict only	UK site presents GBP only

```
{
  "policy_id": "pol_mer_123_v4",
  "mode": "allowlist",
  "default": ["USD"],
  "rules": [
    {"scope": {"channel": "ecommerce"}, "allow": ["USD", "CAD", "EUR", "GBP", "JPY"]},
    {"scope": {"channel": "pos"}, "allow": ["USD"]},
    {"scope": {"mop": "amex"}, "deny": ["JPY"]},
    {"scope": {"storefront": "uk_site"}, "allow": ["GBP"]}
  ],
  "on_unsupported": "reject",
  "effective_from": "2026-09-21T00:00:00Z"
}
```

Policy rules:

- **Restrict only.** A policy entry outside platform capability is rejected at save time with the failing dimension named. It never silently widens anything.
- **Explicit allowlist, not default-all.** New merchants start with their home currency. Defaulting to every capable currency would expose merchants to presentment they have not priced for and FX on payout they did not expect.
- **Widening still triggers setup checks.** Adding a currency to the policy runs the payout-path and settlement-mapping checks from setup. Narrowing runs none.
- **Most specific scope wins, deny beats allow.** Storefront beats channel, channel beats merchant default. A deny at any level removes the currency.

- **Policy is versioned.** Each Payment records the policy version with the capability snapshot version. Narrowing the policy never blocks captures, refunds or disputes on existing Payments.
- **No FX fallback.** If a requested currency is outside the effective set, the Session is rejected. The platform never converts the price into an allowed currency on the merchant's behalf; merchants who want that use a pricing or localization feature upstream of the Session.

Merchant currency strategy

The policy above is governance: which currencies this merchant may use. The strategy is optimization: how the platform should use them. They are separate objects with separate change cadences and separate approvers.

Control is two independent axes rather than a list of named modes, because the two questions are independent and named modes multiply every time an option is added:

- `presentment_control`: `merchant` or `platform_recommended` — who decides the currency the shopper is priced in.
- `path_control`: `merchant_directed` or `platform_optimized` — who decides the acquiring entity, network, settlement target and FX timing behind it.

Three presets cover the merchant-facing conversation:

Preset	presentment_control	path_control	Fit
Merchant-controlled	merchant	merchant_directed	Runs its own interchange model and treasury
Hybrid (recommended default)	merchant	platform_optimized	Merchant owns the customer-facing price; the platform optimizes entity, network, settlement and FX
Platform-optimized	platform_recommended	platform_optimized	Wants the platform to run currency and routing end to end

```
{
  "currency_strategy": {
    "presentment_control": "merchant",
    "path_control": "platform_optimized",
    "default_presentment": "USD",
    "local_presentment": {"enabled": true, "countries": ["GB", "CA", "AU", "DE", "FR"]},
    "preferred_settlement": ["USD", "EUR"],
    "preferred_payout": ["USD"],
    "economic_objective": {"type": "blended", "weights": {"cost": 0.6, "auth_rate": 0.4}},
    "refund_fx_policy": "merchant_bears",
    "substitution": {"enabled": true, "order": ["EUR", "GBP", "USD"], "applies_at": "quote"},
    "strategy_version": "cst_mer_123_v2"
  }
}
```

The economic objective here is the same objective the routing modes use. One vocabulary, not two, so a merchant cannot configure FX to minimize cost while routing maximizes auth rate and neither side knows which won.

Substitution. Silent substitution stays prohibited. A merchant may configure an explicit fallback order, and it applies only where the shopper still sees and accepts the final currency before paying, or at a recurring contract change with advance notice. It never applies at authorization, where the shopper has already agreed to an amount — that is the DCC boundary, not a fallback.

Routing control modes

Merchants split into two types, and the difference is a per-merchant setting, not a bespoke integration: one type wants the network order fixed and reproducible, the other wants the platform to choose per transaction. Both modes are evaluated only after eligibility filtering, and neither can change the transaction currency.

	Merchant-directed	Platform-optimized
Who picks the network per Attempt	The merchant's fixed preference order	The platform, per Attempt
Objective	Whatever the merchant encoded	Merchant-selected: lowest cost, highest auth rate, or blended
Failover	Only to alternates the merchant listed, and only on ineligibility or network unavailability	Any eligible network, per routing policy, on network failure or timeout
Retry after decline	Off unless the merchant enables it, within listed alternates and scheme limits	Soft declines only, within scheme reattempt limits; hard declines never
New network the platform certifies	Unused until the merchant adds it	Eligible automatically once certified
Reporting	Shadow report: what optimization would have chosen	Realized uplift against the merchant's static order
Accountability	Merchant owns the cost and auth-rate outcome	Platform reports against the merchant's stated objective
Typical merchant	Runs its own interchange model, or has scheme volume commitments	Wants the platform to run routing and prove the result

Guardrails that bind both modes. Eligibility comes first: the currency × MOP × network × acquiring entity matrix decides the candidate set, so a network that cannot process the currency is never selected in either mode. Cartes Bancaires on a GBP presentment is the worked example: CB is EUR-only, so it leaves the candidate set and the co-badge partner takes the transaction. Reg II unaffiliated-network availability, IFR Article 8 shopper choice, scheme rules and compliance exclusions also bind both modes, and the presentment-equals-submission invariant means no mode may convert currency to make a preferred network eligible. Two further constraints narrow the candidate set before either mode is applied: a network token is bound to its brand, so tokenized transactions can only be re-routed where PAR or the debit network's token service exists, and a network whose settlement currency has no payout path for this merchant is not a candidate at all.

Both modes can be scoped by channel, storefront, MOP and BIN class, so a platform-optimized merchant can still pin one segment, and a merchant-directed merchant can hand over one segment.

```
{
  "routing": {
    "mode": "merchant_directed",
    "preference": [
      {"scope": {"mop": "visa", "issuer_country": "FR", "currency": "EUR"},
        "order": ["cartes_bancaires", "visanet"]},
      {"scope": {"mop": "visa"}, "order": ["visanet"]}
    ],
    "fallback_on_ineligible": "listed_alternates_only",
    "on_no_eligible_network": "decline_and_alert",
    "routing_policy_version": "rt_mer_123_v7"
  }
}
```

```
{
  "routing": {
    "mode": "platform_optimized",
    "objective": "blended",
    "objective_weights": {"cost": 0.6, "auth_rate": 0.4},
    "pinned": [
      {"scope": {"mop": "us_debit", "bin_class": "regulated"}, "prefer": ["star", "nyce"]}
    ],
    "excluded_networks": [],
    "retry_on_decline": {"enabled": true, "max_attempts": 2},
    "fallback_on_ineligible": "next_eligible",
    "routing_policy_version": "rt_mer_456_v3"
  }
}
```

The routing policy is versioned and recorded on each Payment alongside the capability snapshot and currency policy versions, so a mode or objective change never re-decides a transaction that already ran.

Platform-supported presentment and transaction currencies

Launch with about 40 presentment currencies in three tiers, not the full network list. Stripe advertises 135+ ([Stripe](#)) and Adyen lists about 130 codes ([Adyen](#)), but breadth only matters where acquiring configuration, ledger handling and settlement liquidity exist behind it.

Tier membership

A tier is a statement about operational maturity at an acquiring entity, not a permanent property of a currency. A currency moves up as the lifecycle behind it is completed, and the same currency can sit at different tiers on different entities.

- **Tier 1, fully native:** presentment, submission, settlement, balance, payout, refund, dispute, FX corridor and reconciliation all supported end to end. At launch: USD, CAD, EUR, GBP, AUD, NZD, JPY, CHF, SEK, NOK, DKK, PLN, HKD, SGD, MXN.
- **Tier 2, transaction supported:** processing works, with documented restrictions on settlement, payout, balance retention, refund FX or local acquiring. At launch: AED, SAR, QAR, ILS, CZK, HUF, RON, ZAR, THB, MYR, PHP, TWD, KRW, NGN, KES, COP, PEN, EGP.
- **Tier 3, controlled:** requires a local entity, special certification, compliance review, a named partner or special treasury handling. Three-decimal currencies, exponent deviants and FX-controlled currencies sit here until each stage is proven.
- **Excluded:** currencies of comprehensively sanctioned jurisdictions for a US platform, such as CUP, plus RUB and BYN pending sanctions review. Adyen lists all three, which shows a vendor list is not a compliance list.

Each currency's tier, entity and effective date live in the capability graph, so promotion is a recorded change rather than an edit to a static list.

Currencies needing special handling

Currency	ISO exponent	Adyen API exponent	Stripe handling	Platform rule
JPY	0	0	Zero-decimal; 50 JPY minimum	Reject fractional input; exponent 0 end to end
KRW	0	0	Zero-decimal; 50 KRW minimum	Same as JPY
VND, PYG, XOF, XAF	0	0	Zero-decimal	Same as JPY
UGX	0	0	Two-decimal value, fraction must be 00	Store exponent 0; convert per adapter
ISK	0	2	Two-decimal value, fraction must be 00	Store exponent 0; adapter pads to 2 where the network or partner requires
CLP	0	2	Not a listed special case	Store exponent 0; per-network wire exponent from certification

Currency	ISO exponent	Adyen API exponent	Stripe handling	Platform rule
IDR	2	0	0.50 IDR minimum listed	Store exponent 2; reject nonzero fraction until network wire format is certified
CVE	2	0	Not listed	Tier 3; certify before enabling
HUF	2	2	Charges 2-decimal; payouts integer only	Payout engine rounds to whole units; residual stays on balance
TWD	2	2	Charges 2-decimal; payouts integer only	Same as HUF
BHD, KWD, JOD, OMR, TND	3	3	Not in general support list	Tier 3; confirm each network's three-decimal clearing support
BRL, INR, ARS	2	2	Subject to FX-control remittance rules	Local acquiring entity or licensed partner only; no cross-border presentment by default

Adyen's exponent deviations for CLP, CVE, IDR and ISK and Stripe's ISK, UGX, HUF and TWD rules come from their currency pages ([Adyen](#), [Stripe](#)). Network wire exponents must come from the platform's own Visa, Mastercard, Amex and Discover certification, not from either vendor.

Amount limits

Payment, settlement and payout minimums are three different thresholds and should not be collapsed into one check. The payment minimum is whatever the network or acquirer enforces, in the currency it enforces it in; settlement and payout minimums belong to their own stages, where a threshold holds a payout rather than rejecting an authorized payment. Stripe applies its minimum after conversion to the default settlement currency, for example 0.50 USD, 0.50 EUR, 0.30 GBP and 50 JPY. Maximums differ by network: Stripe allows up to 12 digits in minor units for most cards but 9 digits for Amex, and 8 digits for JCB, Diners and Discover on Japanese accounts ([Stripe](#)).

Currency capability graph and corridors

The capability matrix answers whether the platform can process EUR. The graph answers the question that decides whether a merchant gets paid: how money moves from this shopper to this merchant's bank account. Each edge is a path, and it exists only when every hop is configured and certified.

An edge is entity × MOP × network × presentment currency × settlement currency × payout currency, with an effective date. Two paths that share a presentment currency are different products:

Path	Where FX happens	Who carries the currency risk before payout	Fits
EUR presentment → EUR submission → EUR settlement → EUR balance → USD payout	At payout, on the merchant's schedule	Merchant, while the EUR balance is held	Merchants with EUR costs, EUR refunds or a EUR treasury view
EUR presentment → EUR submission → USD settlement → USD payout	At settlement, per transaction	Platform rate applied transaction by transaction	Merchants who want USD only and no balance to manage

A corridor is the FX-capable edge between two points the graph already connects.

```
{
  "corridor_id": "cor_eur_usd_01",
  "from": "EUR",
  "to": "USD",
  "purpose": "settlement_to_payout",
  "entity": "jpmc_us_01",
  "rate_source": "platform_treasury",
  "markup_bps": 100,
  "liquidity_tier": "deep",
  "constraints": ["no_fx_control"],
  "effective_from": "2026-10-01",
  "status": "active"
}
```

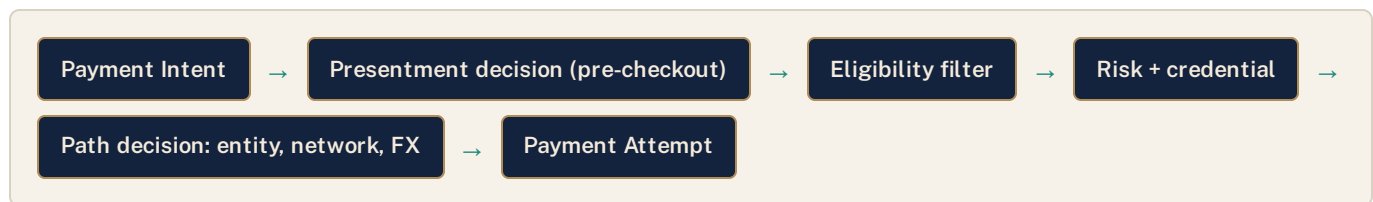
Corridors are materialized from entity, scheme and treasury configuration, never maintained in parallel with it. The settlement mapping stays the single source of truth for where a transaction settles; a corridor only prices conversion between two points that already exist, so the two cannot drift apart. A currency with no corridor to one of the merchant's payout currencies is not eligible for that merchant, however well the networks support it.

Currency decision layer

Currency decisions happen at two points, and keeping them apart is what stops optimization from becoming unauthorized conversion.

1. The presentment decision, before the shopper sees a price. Inputs: shopper country, issuer country where known, channel, merchant strategy, and the effective set from capability $\cap$ policy. Output: the currency the storefront prices in. The platform recommends it, or selects it where the merchant has delegated that. Either way the shopper sees and agrees to that currency before consent.

2. The path decision, after presentment is fixed. Inputs: eligibility, credential and token constraints, regulatory constraints, the auth-rate model, the cost model, corridor FX cost and treasury preference. Output: acquiring entity, network, settlement target where more than one configuration exists, FX timing and payout corridor. It may never change presentment or submission currency.



What is actually choosable. Settlement currency is not a free per-transaction variable. It follows the acquiring entity and BIN configuration and the scheme's own rules, so the real lever is which entity and BIN the transaction is acquired on. Modeling it as "choose a settlement currency" would promise merchants something the schemes do not permit.

One economic objective. Routing, entity selection and FX timing are scored against the same merchant objective — lowest cost, highest auth rate or a weighted blend — evaluated in the platform's economic currency. That currency is an internal unit for comparing options: it never posts to a ledger, never appears as a merchant-facing amount, and never touches what the shopper pays.

Explainability is not optional. Every decision records the candidate set, the chosen path, the runner-up, the deciding factor and the model and configuration versions used. For US debit, least-cost selection runs only on recorded merchant authorization. A score that determines merchant economics has to be explainable on request; an opaque internal score inside a bank-owned PSP is a supervisory question waiting to be asked.

Treasury, balances and FX

FX is where the money is and where the disputes start, so quotes, balances and exposure are objects rather than fields on a ledger row.

Balances. One per merchant per currency, with states: available, pending, reserved (risk holds), payable (net of fees and scheduled for payout) and exposure (unrealized FX on open refund and dispute liability). Balances hold the settlement currency until payout, so conversion happens once.

```
{
  "quote_id": "fxq_123",
  "party": "merchant",
  "source_currency": "EUR",
  "target_currency": "USD",
  "source_amount": 100000,
  "reference_rate": 1.1724,
  "applied_rate": 1.1841,
  "markup_bps": 100,
  "rate_source": "platform_treasury",
  "corridor_id": "cor_eur_usd_01",
  "rate_timestamp": "2026-09-24T18:02:11Z",
  "expires_at": "2026-09-24T18:32:11Z",
  "locked": true
}
```

Every converted Payment, refund, dispute adjustment and Payout references a quote id. That gives reproducibility, disputeability, revenue attribution with markup separated from the reference rate, rate locking and clean reconciliation.

Exposure is an entry, not a rounding difference. Capture EUR 1,000 settling at 1.1000 is USD 1,100. The same EUR 1,000 refunded thirty days later at 1.0700 costs USD 1,070, leaving USD 30 unaccounted. The ledger posts the original settlement, the revaluation at the refund-date rate, the FX gain or loss, and the liability allocation to merchant or platform per the refund FX policy, rather than absorbing it into a difference line nobody owns.

Customer-side and merchant-side FX never mix. DCC quotes and payout FX quotes share this schema — duplicating it would breed drift — but they are separate instances with a `party` discriminator, separate disclosure, separate markup caps and separate accounting. DCC margin never posts to merchant FX revenue, and a merchant payout quote is never shown to a cardholder.

Treasury owns position limits. Holding EUR balances is a position. Per-currency limits, and the corridor liquidity behind them, belong to treasury, not to the routing model.

Recurring billing and multi-currency orders

Subscriptions carry currency explicitly: contract currency (what was agreed), billing currency (what the invoice is raised in) and payment currency (what the shopper is charged). Settlement and payout follow the path decision as usual.

History is immutable. A currency change never rewrites past invoices or payments. They stay in the currency they were agreed and charged in, which is what refunds and disputes reconcile against.

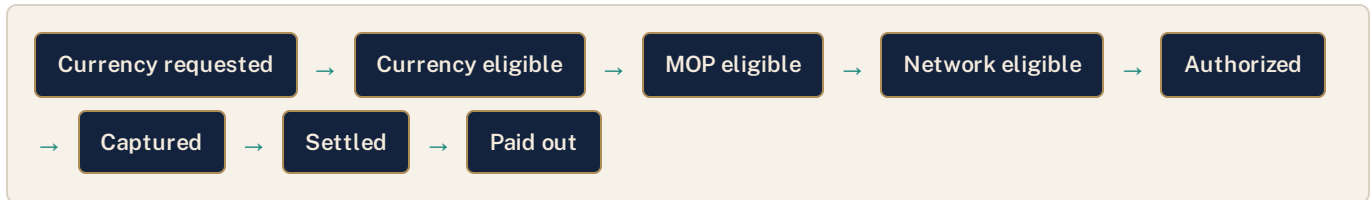
Migration is a policy, not a side effect. When a customer moves country, the merchant's strategy decides whether to keep the contract currency or bill future periods in the new one. A switch changes what the customer pays, so it needs fresh consent — a new mandate for direct debit, a customer-initiated confirmation for card — plus advance notice. It is never done silently at the next renewal.

Orders. An Order carries an order currency, optional line-item currencies and a canonical valuation currency for reporting. Mixed-currency line items are normal in travel and marketplaces, but a single Order settling in several currencies makes refunds, disputes and reconciliation ambiguous, so multiple payment currencies within one Order stay out of scope until the balance and exposure model above is live.

Reporting and observability

Reporting currency. Finance needs one currency; operations keep several. The reporting currency is a derived view with its own rate source and as-of date, and converting for reporting never mutates a ledger entry. A transaction that settles EUR 1,000 and pays out USD 1,170 can report USD 1,168.42 without either operational figure changing.

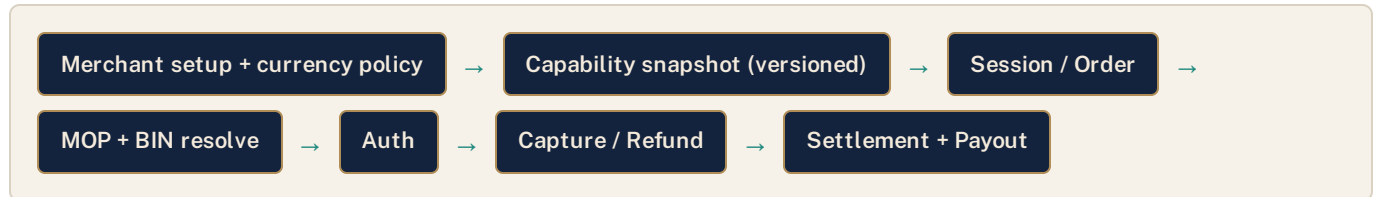
The currency funnel. Each stage carries a reason code, so a loss is attributable rather than inferred.



Beyond the funnel: auth rate by presentment currency and by settlement currency, cross-border share, FX cost, FX revenue and realized markup, refund and dispute FX exposure, currency-related declines, currency-related routing failures, settlement breaks, conversion volume, merchant currency adoption, and profitability per currency and per corridor — which is what decides the next currency worth certifying.

Validation: setup time vs transaction time

Anything that depends only on the merchant's contract and configuration is validated once at setup and frozen into a versioned capability snapshot. Anything that depends on the shopper, the card or the amount is validated per transaction against that snapshot, with no live calls to contract systems in the auth path.



Setup-time validation (onboarding, MOP enablement, policy or config change)

Check	Rule	On failure
Currency code	Valid ISO 4217 alphabetic code in the platform registry, not excluded	Reject config
Sanctions and FX control	Not a sanctioned currency; controlled currencies (BRL, INR, ARS) require a local entity or licensed partner	Reject, or route to compliance review
MOP × currency	Currency is in the MOP's supported set; Amex checked against the SE contract	Reject that currency for that MOP only
Acquiring entity × currency	Acquiring BIN/ICA is configured for the currency with each network	Reject; open provisioning ticket
Settlement mapping	Every MOP × currency pair resolves to a settlement currency	Reject
Payout path	Each settlement currency converts to a payout account currency, or has its own payout account	Reject, or require an FX election
Ledger support	Exponent registered; three-decimal and deviant currencies certified on each network adapter	Keep currency gated at Tier 3
Merchant policy	Every allowed currency sits inside platform capability; partner policy bounds sub-merchant policy	Reject policy save, naming the failing dimension
Domestic scheme	Cartes Bancaires, eftpos or Interac only with a local entity and domestic currency	Reject
MCC restrictions	Currency and MOP allowed for MCC, such as gambling or crypto	Reject or review
DCC enablement	Network DCC registration, disclosure templates, markup within regulatory caps such as EU CBPR2	Keep DCC disabled
Snapshot publish	Compile all of the above into an immutable, versioned capability snapshot	Previous snapshot stays live

Transaction-time validation

Stage	Check	On failure
Session / Order create	Currency is in the merchant's effective presentment set (capability ∩ policy)	400 <code>currency_not_enabled</code>

Stage	Check	On failure
Session / Order create	Amount is an integer in minor units at the currency's API exponent; zero-fraction rule for ISK, UGX and similar	400 <code>invalid_amount_precision</code>
Session / Order create	Amount at or above the payment minimum, in the currency the network or acquirer enforces it in; payout thresholds are not applied here	400 <code>amount_too_small</code>
Session / Order create	Only MOPs valid for this currency are returned to checkout	MOP hidden, not an error
MOP + BIN resolve	Card brand from BIN or token is enabled; network of record chosen per routing and Reg II or IFR choice	Decline pre-auth with <code>brand_not_enabled</code>
MOP + BIN resolve	Amount within the chosen network's maximum digits, such as 9 digits for Amex	Pre-auth reject <code>amount_too_large</code>
MOP + BIN resolve	Shopper's billing currency differs from presentment: flag for analytics; offer DCC only if enabled	Informational
Auth	Submission currency equals presentment currency unless a DCC quote is attached, accepted and unexpired	Hard block (platform invariant)
Auth	Exponent converted to the network's wire format at the adapter; round-trip check	Hard block
Capture	Currency equals the auth currency; amount within the network's capture tolerance of the auth	Reject capture
Refund	Currency equals the original transaction currency; cumulative refunds at or below captured amount	Reject refund
Dispute	Chargeback and representment in the transaction currency; FX difference posted to the merchant ledger per policy	Ledger exception queue
Settlement	Settlement currency received matches the snapshot mapping for MOP × entity × currency	Reconciliation break
Payout	FX rate and markup recorded on the ledger entry; HUF and TWD payouts rounded to whole units	Hold payout

Snapshot versioning. Each Payment records the capability snapshot version and policy version it was validated against. A later config or policy change never invalidates in-flight captures, refunds or disputes, which are validated against the original snapshot plus the lifecycle rules above.

Requirements

Each requirement follows one pattern: **As a PSP platform, I shall [capability], provided that [preconditions], so that [merchant or platform outcome]**. Acceptance criteria sit under each statement and are written to be testable.

IDs use the prefix REQ-CUR-, grouped by area: CAP (capability, setup, policy and strategy), PRS (presentment and amount), RTE (routing and submission), LFC (lifecycle), STL (settlement, FX, payout), CMP (DCC and compliance), COR (capability graph and corridors), DEC (currency decision), FXQ (FX quotes and exposure), BAL (balances and treasury), REC (recurring billing and orders), RPT (reporting and observability). Priority: **P0** is required for launch, **P1** for Tier 2 expansion, **P2** is roadmap.

Capability matrix, merchant setup and currency policy

1. **REQ-CUR-CAP-01** **P0** As a PSP platform, I shall allow a merchant to enable a presentment currency for a MOP, provided that the currency is supported by every network eligible to process that MOP, the acquiring entity is configured for it with those networks, and the platform ledger supports its exponent, so that the merchant can accept payments in that currency without downstream failures.
 - Enabling a currency with any failing dimension is rejected, and the response names the failing dimension (MOP, network, acquiring entity, ledger).
 - A currency valid for Visa but not Amex is enabled for Visa only, never rejected for the whole merchant.
2. **REQ-CUR-CAP-02** **P0** As a PSP platform, I shall maintain a capability matrix keyed by MOP × network × acquiring entity × currency, provided that each entry is sourced from network certification and acquirer configuration, not vendor documentation, so that currency availability reflects what the platform can actually process.
 - Each matrix entry records its source, effective date and owner.
 - A matrix entry cannot be activated without a linked certification record.
3. **REQ-CUR-CAP-03** **P0** As a PSP platform, I shall require every enabled MOP × currency pair to resolve to a settlement currency and a payout path, provided that the merchant has either a payout account in that currency or an FX election to a payout account currency, so that no captured funds are left without a route to the merchant.
 - Setup is rejected when a pair has no settlement mapping or no payout path.
4. **REQ-CUR-CAP-04** **P0** As a PSP platform, I shall compile validated merchant configuration into an immutable, versioned capability snapshot, provided that all setup checks pass, so that transaction-time validation runs against a consistent, cached view without calling contract systems.
 - A failed compile leaves the previous snapshot live.
 - Snapshot lookup adds no more than 2 ms at p99 to the auth path (target to confirm).
5. **REQ-CUR-CAP-05** **P0** As a PSP platform, I shall enable Amex presentment currencies only from the merchant's Amex acquiring model and SE contract, provided that the model is OptBlue or direct SE, so that the merchant is never offered a currency Amex will not accept for that SE.
6. **REQ-CUR-CAP-06** **P1** As a PSP platform, I shall allow domestic co-badged schemes such as Cartes Bancaires, eftpos and Interac only in their domestic currency, provided that the merchant contracts through a local acquiring entity, so that domestic routing and interchange apply correctly.

7. **REQ-CUR-CAP-07** **P0** As a PSP platform, I shall classify every platform currency into Tier 1, Tier 2, Tier 3 or Excluded, provided that Tier 3 enablement requires a per-merchant review, so that operational risk grows only where support has been proven.
8. **REQ-CUR-CAP-08** **P0** As a PSP platform, I shall let a merchant define a currency policy that restricts presentment currencies by MOP, channel and storefront, provided that every allowed currency is within platform capability, so that merchants accept only the currencies they have priced and planned treasury for.
 - o A policy that lists a currency outside capability is rejected at save, naming the failing dimension.
 - o The effective set at Session create equals capability $\cap$ policy.
9. **REQ-CUR-CAP-09** **P1** As a PSP platform, I shall let a partner or marketplace set a policy that bounds its sub-merchants' policies, provided that the partner operates those sub-merchants, so that platform customers can enforce a consistent currency footprint.
 - o A sub-merchant policy cannot allow a currency its partner policy denies.
10. **REQ-CUR-CAP-10** **P0** As a PSP platform, I shall version every currency policy and record the version on each Payment, provided that the policy changes after the Payment is created, so that narrowing a policy never blocks lifecycle operations on existing Payments.

Presentment, amount and exponent handling

1. **REQ-CUR-PRS-01** **P0** As a PSP platform, I shall accept an Order or Session in a presentment currency, provided that the currency is in the merchant's effective set in the current capability snapshot and policy, so that the shopper is only ever shown prices the platform can process.
 - o A disabled currency returns 400 `currency_not_enabled` before any MOP is offered.
2. **REQ-CUR-PRS-02** **P0** As a PSP platform, I shall return to checkout only the MOPs that support the Order's presentment currency, provided that the MOP is enabled for the merchant, so that the shopper cannot select a MOP that will fail on currency.
 - o Unsupported MOPs are hidden, not shown as errors.
3. **REQ-CUR-PRS-03** **P0** As a PSP platform, I shall accept amounts only as integers in minor units at the currency's platform API exponent, provided that the exponent is stored on the money object itself, so that no amount is misread by a factor of 10, 100 or 1,000.
 - o Every money object carries currency, amount and exponent.
 - o Three-decimal currencies (BHD, KWD, JOD, OMR, TND) round-trip without loss.
4. **REQ-CUR-PRS-04** **P0** As a PSP platform, I shall reject nonzero fractional amounts for currencies represented with padded decimals, such as ISK and UGX, provided that the currency is flagged zero-fraction in the registry, so that the platform never submits a fraction the network or issuer cannot bill.
 - o ISK 5.50 is rejected; ISK 5.00 is accepted.
5. **REQ-CUR-PRS-05** **P0** As a PSP platform, I shall maintain separate ISO, API and network wire exponents per currency, provided that wire exponents come from network certification, so that deviations such as CLP, ISK and IDR are handled at the adapter without leaking into merchant APIs.

6. **REQ-CUR-PRS-06** **P0** As a PSP platform, I shall enforce the payment minimum in the currency in which the network or acquirer actually enforces it, provided that the applicable minimum is loaded into the snapshot, so that a transaction is never priced below its processing cost and is never rejected at checkout for a threshold that belongs downstream.
 - Settlement and payout minimums are separate concepts, evaluated at their own stage (REQ-CUR-BAL-02), and never applied as a payment-time rejection.
7. **REQ-CUR-PRS-07** **P0** As a PSP platform, I shall enforce a maximum amount per network and currency, provided that network digit limits are loaded into the snapshot, so that oversize transactions fail before submission rather than at the network.
 - An Amex amount above 9 digits in minor units is rejected pre-auth.

Network routing and submission

1. **REQ-CUR-RTE-01** **P0** As a PSP platform, I shall submit authorization and clearing in the same currency and amount the shopper agreed to at presentment, provided that no accepted DCC quote is attached, so that the shopper is billed exactly what they approved and the transaction is not disputable on currency grounds.
 - Any Attempt where submission currency differs from presentment without a DCC quote is blocked as a platform invariant, not a configurable rule.
2. **REQ-CUR-RTE-02** **P0** As a PSP platform, I shall select the network of record per Attempt from BIN or token attributes, merchant eligibility and routing policy, provided that the selected network supports the presentment currency in the snapshot, so that a co-badged card never routes to a network that cannot process the currency.
 - When the preferred network lacks the currency, routing falls to the next eligible network or fails with `currency_not_supported_by_network`.
 - A domestic scheme restricted to one currency, such as Cartes Bancaires in EUR, is removed from the candidate set for any other presentment currency and is hidden at checkout rather than failing at auth.
3. **REQ-CUR-RTE-03** **P0** As a PSP platform, I shall keep at least two unaffiliated debit networks enabled and available for US debit transactions, including card-not-present, provided that the issuer supports them and the presentment currency is USD, so that the merchant keeps Reg II routing choice.
 - Least-cost debit routing by the platform runs only on explicit merchant authorization recorded in the routing policy, not on a default toggle.
4. **REQ-CUR-RTE-04** **P1** As a PSP platform, I shall honor shopper brand choice on EU co-badged cards, provided that the merchant is on an EU acquiring entity, so that routing complies with IFR Article 8.
5. **REQ-CUR-RTE-05** **P0** As a PSP platform, I shall route wallet and network-token transactions on the underlying card network with the card MOP's currency rules, provided that the token maps to an enabled card brand, so that Apple Pay, Google Pay and Click to Pay add no separate currency configuration.
 - Tokenization constrains routing: a network token is bound to its brand, so alternative debit routing requires PAR or the debit network's own token service. Where neither exists, the brand rail is the only candidate and the routing decision records that constraint.
6. **REQ-CUR-RTE-06** **P0** As a PSP platform, I shall convert amounts to the network wire exponent at the network adapter and verify a lossless round-trip before sending, provided that the wire exponent is certified, so that encoding errors are caught before they reach the network.
7. **REQ-CUR-RTE-07** **P0** As a PSP platform, I shall record on each Attempt the presentment currency, submission currency, network of record, capability snapshot version, currency policy version and routing policy version, so that every transaction can be audited and reconciled against the rules in force when it ran.

8. **REQ-CUR-RTE-08** **P0** As a PSP platform, I shall let a merchant choose a routing control mode of merchant-directed or platform-optimized, provided that both modes are applied only after eligibility filtering, so that merchants decide how much routing control to hand over without either mode overriding network, regulatory or currency rules.
- The mode and routing policy version are recorded on every Attempt.
 - Neither mode can select a network outside the eligible candidate set, and neither can convert currency to make one eligible.
 - A network is eligible only if its settlement currency maps to a payout path for this merchant (REQ-CUR-CAP-03), so a routing choice can never strand funds.
 - Both modes can be scoped by channel, storefront, MOP and BIN class.
9. **REQ-CUR-RTE-09** **P0** As a PSP platform, I shall follow a merchant-directed merchant's preference order deterministically, provided that the preferred network is eligible for that Attempt, so that the merchant can reproduce and audit every routing decision.
- Given the same inputs and the same eligibility state, the same network is always chosen; when eligibility changes, such as a network outage, the Attempt records what changed.
 - When the preferred network is ineligible, routing falls only to alternates the merchant listed; when none is eligible, the Attempt is declined with `no_eligible_network` and a merchant alert, never silently substituted.
10. **REQ-CUR-RTE-10** **P0** As a PSP platform, I shall select the network per Attempt for a platform-optimized merchant to maximize that merchant's stated objective of lowest cost, highest auth rate or a weighted blend, provided that the objective is configured and every candidate is eligible, so that the merchant gets platform routing without giving up rule compliance or explainability.
- Each Attempt records the chosen network, the runner-up and the deciding factor.
 - Cost decisions record the cost-model version used, and estimated cost is reconciled against actual interchange and scheme fees after clearing, because interchange is finalized at clearing rather than at auth.
 - Auth-rate decisions require a minimum sample per issuer or BIN segment plus an exploration slice, so the model does not starve the alternative network of the data it needs.
 - Objective and weight changes are versioned; in-flight Payments keep the version they ran under.
11. **REQ-CUR-RTE-11** **P0** As a PSP platform, I shall fail over to an alternate eligible network when the chosen network is unavailable or times out, provided that no issuer decision was returned, so that network-level failures do not become lost sales.
- Failover is capped per Attempt and recorded with the failure reason.
12. **REQ-CUR-RTE-12** **P0** As a PSP platform, I shall not reattempt an issuer decline on a different network in order to escape that decline, provided that the decline came from the issuer rather than the network, so that scheme reattempt rules are respected and a hard decline is honored.
- Hard declines such as pickup, stolen card, closed account or failed authentication are never retried on any network.
 - Soft declines are retried only within scheme reattempt limits and the merchant's retry configuration; a same-issuer retry on a different network is treated as unlikely to change the outcome and is off by default.

- 13. REQ-CUR-RTE-13 P1** As a PSP platform, I shall report realized routing outcomes against the alternative, provided that eligibility can be reconstructed from each Attempt's snapshot, so that a merchant can change routing mode on evidence rather than assertion.
- Platform-optimized uplift is measured against an exploration or holdout slice, not against a modeled counterfactual alone.
 - Merchant-directed merchants see a shadow report of what optimization would have chosen, labeled as an estimate, not a guarantee.

Capture, refund and dispute lifecycle

- 1. REQ-CUR-LFC-01 P0** As a PSP platform, I shall capture only in the authorization currency, provided that the capture amount is within the network's tolerance of the authorized amount, so that clearing matches the authorization and avoids scheme integrity fees.
 - A capture in a different currency is rejected; a capture above tolerance is rejected or requires incremental authorization.
- 2. REQ-CUR-LFC-02 P0** As a PSP platform, I shall process refunds only in the original transaction currency, provided that cumulative refunds do not exceed the captured amount in that currency, so that the shopper receives back exactly what they paid, in the currency they paid.
 - A partial refund in a different currency is rejected, even if the merchant requests it.
- 3. REQ-CUR-LFC-03 P0** As a PSP platform, I shall validate lifecycle operations against the Payment's original capability snapshot and policy version, provided that the merchant's configuration has since changed, so that disabling a currency never strands in-flight captures, refunds or disputes.
 - Disabling EUR for a merchant still allows refunds on prior EUR captures.
- 4. REQ-CUR-LFC-04 P0** As a PSP platform, I shall receive chargebacks and submit representments in the transaction currency, provided that the network presents the dispute in that currency, so that dispute amounts reconcile to the original Payment.
- 5. REQ-CUR-LFC-05 P0** As a PSP platform, I shall post the FX difference between capture-time and refund-time or dispute-time rates to the merchant ledger per the merchant's refund FX policy, provided that the payout currency differs from the transaction currency, so that FX exposure is explicit and never absorbed silently.
 - Each FX difference is a separate ledger entry with both rates recorded.
- 6. REQ-CUR-LFC-06 P1** As a PSP platform, I shall let a merchant choose a refund FX policy (merchant bears, platform bears within a window, or rate locked at capture), provided that the policy is contractually approved, so that FX risk allocation is a commercial choice rather than an accident.

Settlement, FX, payout and ledger

- 1. REQ-CUR-STL-01 P0** As a PSP platform, I shall reconcile each network settlement against the expected settlement currency for its MOP × acquiring entity × transaction currency, provided that the mapping exists in the snapshot, so that unexpected settlement currencies surface as reconciliation breaks instead of silent ledger errors.

2. **REQ-CUR-STL-02** **P0** As a PSP platform, I shall convert from settlement currency to payout currency once, at the platform FX rate plus the merchant's contracted markup, provided that the merchant has no payout account in the settlement currency, so that the merchant is paid in their chosen currency with a single, disclosed conversion.
 - o No transaction is converted more than once between settlement and payout.
3. **REQ-CUR-STL-03** **P0** As a PSP platform, I shall record on every FX ledger entry the source currency, target currency, reference rate, applied rate, markup and rate timestamp, so that the merchant can verify FX charges and finance can report FX revenue separately.
4. **REQ-CUR-STL-04** **P0** As a PSP platform, I shall pay out without conversion into a payout account in the settlement currency when one exists, provided that the merchant has enabled that account, so that merchants with multi-currency treasury avoid FX costs.
5. **REQ-CUR-STL-05** **P1** As a PSP platform, I shall apply tiered FX markups by currency tier, provided that tiers are configured in the merchant contract, so that FX pricing reflects liquidity cost for less-traded currencies.
6. **REQ-CUR-STL-06** **P0** As a PSP platform, I shall round payouts to whole units in currencies whose payout rails do not accept fractions, such as HUF and TWD, provided that the residual stays on the merchant balance, so that payouts never fail on precision.
7. **REQ-CUR-STL-07** **P0** As a PSP platform, I shall keep separate merchant balances per currency until payout, provided that the currency is enabled for settlement, so that FX is applied only at the moment funds leave the platform.

DCC, compliance and observability

1. **REQ-CUR-CMP-01** **P2** As a PSP platform, I shall offer DCC as a separate opt-in product that submits in the cardholder's billing currency, provided that the merchant is DCC-registered with the network, the shopper explicitly chooses DCC after seeing the rate and markup, and the quote is unexpired, so that cardholder-side conversion is compliant and never applied by default.
 - o The shopper can always decline DCC and pay in the presentment currency.
 - o Disclosure meets network rules and, for EEA shoppers, CBPR2 markup disclosure.
2. **REQ-CUR-CMP-02** **P0** As a PSP platform, I shall block presentment, settlement and payout in excluded currencies, provided that the exclusion list is owned by compliance and loaded into every snapshot, so that the platform cannot process in sanctioned jurisdictions' currencies.
 - o A list change propagates to live snapshots within a defined SLA (to confirm with compliance).
3. **REQ-CUR-CMP-03** **P0** As a PSP platform, I shall allow controlled currencies such as BRL, INR and ARS only through a local acquiring entity or licensed partner, provided that remittance runs through authorized channels, so that FX-control regulations are met.
4. **REQ-CUR-CMP-04** **P0** As a PSP platform, I shall return machine-readable error codes for every currency rejection, naming the failing check, so that merchants and support can fix configuration without escalation.
5. **REQ-CUR-CMP-05** **P1** As a PSP platform, I shall report authorization rate, cross-border rate and issuer currency mismatch by presentment currency and network, provided that the data is available from the Attempt record, so that merchants can decide which currencies to localize.
6. **REQ-CUR-CMP-06** **P0** As a PSP platform, I shall keep an audit trail of every capability matrix, merchant configuration and currency policy change, with actor, reason and effective time, so that currency availability can be explained to auditors and schemes.

Capability graph and corridors

1. **REQ-CUR-COR-01** **P0** As a PSP platform, I shall model currency capability as a graph of entity × MOP × network × presentment × settlement × payout edges with effective dates, provided that an edge exists only when every hop is configured and certified, so that the platform answers how money moves rather than only whether a currency exists.
 - A currency with no corridor to one of the merchant's payout currencies is not eligible for that merchant.
2. **REQ-CUR-COR-02** **P0** As a PSP platform, I shall maintain corridors as first-class objects carrying purpose, entity, rate source, markup, liquidity and effective dates, provided that they are materialized from entity, scheme and treasury configuration rather than maintained beside it, so that corridor and settlement mapping cannot drift apart.
3. **REQ-CUR-COR-03** **P1** As a PSP platform, I shall version every graph and corridor change and retain superseded versions, so that a historical Payment can be re-evaluated against the graph that was in force when it ran.
4. **REQ-CUR-COR-04** **P1** As a PSP platform, I shall assign each currency an operational maturity tier per acquiring entity, provided that the tier is derived from which lifecycle stages are supported end to end, so that tiering reflects proven capability and the same currency can sit at different tiers on different entities.

Currency decision

1. **REQ-CUR-DEC-01** **P0** As a PSP platform, I shall separate the presentment decision, taken before pricing, from the path decision, taken after presentment is fixed, so that optimization can never change what the shopper agreed to pay.
 - The path decision cannot alter presentment or submission currency.
 - The shopper always sees the final currency before consent, whoever selected it.
2. **REQ-CUR-DEC-02** **P0** As a PSP platform, I shall recommend or select a presentment currency from shopper country, issuer country, channel and merchant strategy within capability n policy, provided that selection happens only where the merchant delegated it, so that local-currency acceptance is a decision with inputs rather than a guess.
3. **REQ-CUR-DEC-03** **P0** As a PSP platform, I shall treat settlement currency as a consequence of acquiring entity and BIN configuration rather than a free per-transaction variable, so that the platform optimizes the lever it actually has, which is entity and BIN selection, and does not promise a settlement choice the schemes do not permit.
4. **REQ-CUR-DEC-04** **P0** As a PSP platform, I shall score routing, entity selection and FX timing against a single merchant economic objective evaluated in the platform's economic currency, provided that the economic currency stays internal to the decision model, so that FX and routing cannot be configured into contradiction and no internal unit reaches a ledger or a merchant-facing amount.
5. **REQ-CUR-DEC-05** **P0** As a PSP platform, I shall record for each currency decision the candidate set, the chosen path, the runner-up, the deciding factor and every model and configuration version used, provided that the decision affects merchant economics, so that it can be explained to a merchant, an auditor or a regulator on request.

FX quotes and exposure

1. **REQ-CUR-FXQ-01** **P0** As a PSP platform, I shall issue an FX quote object for every conversion, carrying party, source and target currency, source amount, reference rate, applied rate, markup, rate source, corridor, timestamp, expiry and lock state, so that every converted amount is reproducible after the fact.
2. **REQ-CUR-FXQ-02** **P0** As a PSP platform, I shall reference the quote id from the Payment, refund, dispute adjustment and Payout it priced, so that reconciliation and FX revenue attribution work from the quote rather than from a rate recomputed later.
3. **REQ-CUR-FXQ-03** **P0** As a PSP platform, I shall keep cardholder-side and merchant-side FX in separate quote instances with separate disclosure, markup caps and accounting, provided that both may share the quote schema, so that DCC margin never posts to merchant FX revenue and no merchant payout quote is ever shown to a cardholder.
4. **REQ-CUR-FXQ-04** **P0** As a PSP platform, I shall post refund and dispute FX differences as explicit ledger entries covering the original settlement, the revaluation at the later rate, the FX gain or loss and the liability allocation under the merchant's refund FX policy, so that exposure is accounted rather than absorbed.
5. **REQ-CUR-FXQ-05** **P1** As a PSP platform, I shall honor a rate lock for the window the merchant contracted, provided that the quote is unexpired, so that a locked refund rate is applied and an expired lock is re-quoted rather than silently extended.

Balances and treasury

1. **REQ-CUR-BAL-01** **P0** As a PSP platform, I shall maintain a per-merchant, per-currency balance with available, pending, reserved, payable and exposure states, so that refunds, reserves, negative balances, disputes and multi-currency payouts have an accounted home.
2. **REQ-CUR-BAL-02** **P0** As a PSP platform, I shall evaluate payment, settlement and payout minimums separately, each at its own stage and in the currency in which it is actually enforced, so that a downstream payout threshold never rejects a transaction the network would have authorized.
 - A payment above the network or acquirer minimum is authorized and captured even when it sits below a payout threshold; the payout is held until the balance clears that threshold.
3. **REQ-CUR-BAL-03** **P1** As a PSP platform, I shall enforce treasury position limits per currency, provided that limits and corridor liquidity are set by treasury, so that how long balances are held is a treasury decision rather than an artifact of routing.

Recurring billing and orders

1. **REQ-CUR-REC-01** **P0** As a PSP platform, I shall carry contract currency, billing currency and payment currency on a subscription, so that a currency change is explicit rather than inferred from the latest charge.
2. **REQ-CUR-REC-02** **P0** As a PSP platform, I shall keep historical invoices and payments immutable in their original currency, so that refunds and disputes reconcile against what was agreed and charged.
3. **REQ-CUR-REC-03** **P1** As a PSP platform, I shall change a subscription's billing currency only through an explicit policy with fresh consent and advance notice, provided that consent matches the instrument (a new mandate for direct debit, a customer-initiated confirmation for card), so that a migration never silently changes what a customer pays.

4. **REQ-CUR-REC-04** **P1** As a PSP platform, I shall carry an order currency, optional line-item currencies and a canonical valuation currency on an Order, provided that multiple payment currencies within one Order stay out of scope until balances and exposure are live, so that reporting is coherent without making refunds and disputes ambiguous.

Reporting and observability

1. **REQ-CUR-RPT-01** **P0** As a PSP platform, I shall express a reporting currency view with its own rate source and as-of date, provided that the conversion never mutates a ledger entry, so that finance sees one currency while operations keep several.
2. **REQ-CUR-RPT-02** **P1** As a PSP platform, I shall instrument the currency funnel from requested through paid out with a reason code at every drop-off, so that currency-driven losses are attributable rather than inferred.
3. **REQ-CUR-RPT-03** **P1** As a PSP platform, I shall report auth rate by presentment and settlement currency, cross-border share, FX cost and revenue, refund and dispute exposure, currency-related declines and routing failures, settlement breaks, and profitability per currency and corridor, extending REQ-CUR-CMP-05, so that the next currency to certify is an evidence decision.

Risks and decisions required

The biggest risk is refund and dispute FX exposure, because the design keeps the shopper whole in the original currency and someone must absorb rate movement between capture and refund.

- **Refund FX policy:** merchant bears it (Stripe-style default), platform bears it within a window, or rate locked at capture for N days.
- **Corridor and position ownership:** whether treasury or payments product owns corridor configuration, markup and per-currency position limits, since routing economics depend on both.
- **Economic objective governance:** who sets the default objective, whether a merchant may see the score behind a decision, and what is disclosed in contract terms rather than on request.
- **Default control preset for new merchants:** hybrid (merchant sets presentment, platform optimizes the path) or fully merchant-controlled until they opt in.
- **Recurring currency migration:** the consent and notice model per region for changing a subscription's billing currency, and whether existing mandates survive it.
- **Mixed-currency Orders:** what has to be true of balances and exposure before multiple payment currencies in one Order move from out-of-scope to supported.
- **Exploration budget:** how much traffic platform-optimized routing may spend testing the alternative network to keep auth-rate models current, and who absorbs the worse outcomes on that slice.
- **Cost-model accuracy:** how often estimated interchange is reconciled against actual post-clearing cost, and what error tolerance forces a model refresh before cost-based routing is trusted.
- **Shadow routing reports:** useful to show merchant-directed merchants what they are leaving on the table, but confirm with legal that an estimate is not read as a commitment.
- **Policy change vs open sessions:** when a merchant narrows its policy, decide whether auth re-checks the current policy or honors the policy version in force when the Session was created.
- **Amex acquiring model:** OptBlue gives the platform control of currency and settlement; direct SE leaves Amex's currency set and settlement outside the platform.
- **Non-US acquiring entities:** JCB, UnionPay, domestic schemes and non-USD Discover presentment depend on which EU/UK and APAC entities are funded.
- **Wire exponent certification:** confirm per-network handling of CLP, ISK, IDR, UGX and three-decimal currencies before any Tier 3 enablement.
- **DCC as a product:** revenue upside against scheme scrutiny, CBPR2 disclosure and conversion risk.
- **Sanctions list ownership:** compliance owns the excluded-currency list; engineering consumes it as snapshot input.

Cross-border fees are not removed by local-currency presentment. Stripe notes issuers may charge the shopper when merchant and card countries differ, regardless of currency ([Stripe](#)). Local acquiring is the only lever for that.

Sources

- [Adyen: Currency codes and minor units](#) (accessed 21 Sep 2026)

- [Stripe: Supported currencies](#) (accessed 21 Sep 2026)